

Evaluation of Wound Healing Activity of *Lantana camara* L. – A Preclinical Study

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***Lantana camara* is used in herbal medicine for the treatment of skin itches, as an antiseptic for wounds, and externally for leprosy and scabies. The objective of our study was to investigate excision wound healing activity of the leaf extract of *L. camara* in rats. The animals were divided into two groups of 12 each in both the models. The test group animals were treated with the aqueous extract of *L. camara* (100 mg/kg/day) topically and the control group animals were left untreated. Wound healing efficacy was measured by determining the morphological and biochemical parameters. Wound healing time, wound contraction and synthesis of collagen were monitored periodically. Antimicrobial activities of the extract against the microorganisms were also assessed. Treatment of the wounds with extract enhanced significantly the rate of wound contraction (98%), synthesis of collagen and decreased mean wound healing time. These studies demonstrate that *L. camara* is effective in healing excision wounds in the experimental animal and could be evaluated as a therapeutic agent in tissue repair processes associated with skin injuries. Copyright © 2008 John Wiley & Sons, Ltd.**

Keywords: *Lantana camara*; hydroxyproline; epithelialization; excision wound.

INTRODUCTION

Wound healing is the body's natural process of regenerating dermal and epidermal tissue. When an individual is wounded, a set of events takes place in a predictable fashion to repair the damage. Following injury, an inflammatory response occurs and the cells below the dermis begin to increase collagen production. Later, the epithelial tissue is regenerated (Iba *et al.*, 2004). There are three stages to the process of wound healing: inflammation, proliferation and remodeling. The proliferative phase is characterized by angiogenesis, collagen deposition, epithelialization and wound contraction. Angiogenesis involves new blood vessel growth from endothelial cells. Subsequently epithelial cells crawl across the wound bed to cover it and the wound is contracted by myofibroblasts, which grip the wound edges and undergo contraction using a mechanism similar to that in smooth muscle cells.

Lantana camara is a significant weed of which there are some 650 varieties in over 60 countries or island groups. *L. camara* is a low erect or subscandent, vigorous shrub with stout recurved prickles and a strong odour of blackcurrants; *L. camara* has several uses, mainly as a herbal medicine and in some areas as firewood and mulch (Sharma *et al.*, 1988, 1999; Day *et al.*, 2003). There has been much work conducted, especially in India, on the chemical constituents of lantana; extracts from the leaves exhibit antimicrobial, fungicidal, insecti-

cidal and nematocidal activity (Begun *et al.*, 1995; Sharma *et al.*, 1999; Day *et al.*, 2003). The use of lantana extracts as potential biocides has been suggested. Verbascoside, which possesses antimicrobial, immunosuppressive and antitumor activities, has been isolated (Day *et al.*, 2003). Lantanoside, linaroside and camarinic acid have been isolated and are being investigated as potential nematocides (Day *et al.*, 2003; Begun *et al.*, 1995). Lantana oil is sometimes used for the treatment of skin itches, as an antiseptic for wounds (Saxena and Sharma, 1999) and externally for leprosy and scabies (Ghisalberti, 2000). Plant extracts are used in folk medicine for the treatment of cancers, chicken pox, measles, asthma, ulcers, swellings, eczema, tumors, high blood pressure, bilious fevers, catarrhal infections, tetanus, rheumatism, malaria and atony of abdominal viscera (Day *et al.*, 2003; Begun *et al.*, 1995). However, there are no scientifically proven data to show whether the *L. camara* has wound healing activity or not. Therefore, the present study was undertaken to explore the effects of *L. camara* leaf extract on excision wound.

MATERIALS AND METHODS

Plant material and extract preparation. The leaves of *L. camara* were collected locally in March 2007 and identified by the plant taxonomist and curator, National Herbarium of Trinidad and Tobago, The University of the West Indies, St Augustine, Trinidad and a voucher specimen was deposited at the above mentioned herbarium (specimen number: 36519). The fresh leaves were shade dried and ground into a powder using an electric blender. The fine powder (100 g) was suspended

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in 200 mL of ethanol for 20 h at room temperature. The mixture was filtered using a fine cloth followed by filter paper (Whatman No 1) and the filtrate was placed in an oven to dry at 40 °C. The dried filtrate (20 g) was used for the study. The extract was subjected to preliminary phytochemical tests.

Animals. The study was approved by the Ethics Committee for animal experimentation (AHC06/07/1), The Faculty of Medical Sciences, The University of the West Indies, St Augustine. An acute toxicity study was conducted to select the dose. The male rats were applied with increasing doses (0.5, 1, 2 and 4 g/kg body weight) of extract for 14 days. The doses up to 4 g/kg body weight did not produce any sign of toxicity and mortality. The animals were physically active and were consuming food and water in a regular way. This was done to determine the toxic effect, if any, of the extract topically.

Healthy inbred male Sprague Dawley rats weighing 200–220 g were used for the study. They were housed individually and maintained on normal food and water *ad libitum*. The animals were periodically weighed before and after the experiment. The rats were anaesthetized prior to and during infliction of the experimental wounds. The surgical interventions were carried out under sterile conditions using ketamine anaesthesia (80 mg/kg body weight). Animals were observed closely for any infection and if they showed signs of infection were separated, excluded from the study and replaced.

Wound-healing activity

Excision wound model. The animals were randomly divided into two groups with a sample size of 12 rats/group in the beginning. Animals were anaesthetized prior to and during creation of the wounds. The rats were inflicted with excision wounds as described by Morton and Malone (1972). The dorsal fur of the animals was shaved with an electric clipper and the anticipated area of the wound to be created was outlined on the back of the animals with methylene blue using a circular plastic cup. A full thickness excision wound with a circular area of 200 mm² and 2 mm depth was created along the markings using toothed forceps, a surgical blade and pointed scissors.

Group 1 (control group) animals were left untreated.

Group 2 (test group) animals were treated topically with extract (100 mg/kg daily) until complete epithelialization. Depending on the staircase experimental observations the lowest dose of 100 mg/kg body weight was used for the treatment group. This treatment was done every day and special care was taken to avoid variation in the dose.

Wound contraction. The wound closure rate was assessed by tracing the wound on alternate days using transparency paper and a permanent marker. The wound areas recorded were measured using graph paper.

Mean clinical epithelization time. All six animals in each group were monitored until complete epithelization (point at which the eschar fell off without any residual raw wound) and the day on which each wound healed was recorded. The mean epithelization time was determined and expressed in days.

Biochemical and histological studies

On day 10 six animals were killed using an overdose of ketamine. The wounds were excised, leaving a 5 mm margin of normal skin around the edges of the wound. The wet weight of the granulation tissue was noted. These granulation tissues were dried at 60 °C for 12 h, weighed and the dry granulation tissue weight recorded. To the dried tissue 5 mL 6 N HCl was added and kept at 110 °C for 24 h. The neutralized acid hydrolysate of the dry tissue was used for the determination of hydroxyproline. An additional piece of wet granulation tissue was preserved in 10% formalin for histological studies. For the better appreciation of collagen deposition, Masson's trichrome stain was used which stains the fibres green.

Estimation of hydroxyproline. Dry granulation tissue from both control and treated groups was used for the estimation of hydroxyproline. Hydroxyproline present in the neutralized acid hydrolysate was subsequently oxidized by sodium peroxide in the presence of copper sulphate. Then it was complexed with paradimethylaminobezaldehyde to develop a pink colour and measured at 540 nm by a spectrophotometer (Neuman and Logan, 1950).

Phytochemical screening methods

Test for saponins. The extract (300 mg) was boiled with 5 mL water for 2 min; the mixture was cooled and mixed vigorously and left for 3 min. The formation of frothing indicates the presence of saponins (Kapoor *et al.*, 1969).

Test for tannins. To an aliquot of the extract (dissolved in water) 2 mL of sodium chloride (2%) was added, filtered and mixed with 5 mL 1% gelatin solution. Precipitation indicates the presence of tannins (Kapoor *et al.*, 1969).

Test for triterpenes. Extract (300 mg) was mixed with 5 mL chloroform and warmed for 30 min. A few drops of concentrated sulphuric acid was added and mixed well. The appearance of red colour indicates the presence of triterpenes (Harborne, 1973).

Test for alkaloids. Extract (300 mg) was digested with 2 M HCl, and the acidic filtrate was mixed with amyl alcohol at room temperature. A pink colour in the alcohol layer indicates the presence of alkaloids (Harborne, 1973).

Test for flavonoids. Extract (300 mg) was boiled with 10 mL ethyl acetate over a steam bath for three minutes. Filtrate of about 4 mL was mixed with 1 mL of dilute ammonia solution. Presence of yellow precipitate indicates the presence of flavonoids (Kapoor *et al.*, 1969).

Antimicrobial activity

Escherichia coli (ATCC 5922), *Klebsiella pneumoniae* (ATCC 700603), *Staphylococcus aureus* (ATCC 4827) and *Salmonella choleraesuis* (ATCC 14028) were the organisms tested. Discs (5 mm) of Whatman filter

paper no. 4 were immersed in *L. camara* leaf extract slurry made using sterile water. The discs were then placed onto the surface of nutrient agar plates inoculated with 24 h cultures of the appropriate organism and scored for the presence of a zone of clearing (positive activity) around the disc over a 48 h period after incubation at 35 °C. Results were analysed visually on the basis of turbid zone of inhibition (+, bacterium colonies deposited in the bottom of the well; ++, turbidity with bacterium).

Data analysis. The means of wound area measurements, between the test and control groups were compared using one-way ANOVA descriptive tests. The epithelization period, weight and hydroxyproline content of the granulation tissue between the test and control groups were compared using independent *t*-tests. Data were analysed using SPSS (Version 12.0, Chicago, USA) and a value of $p < 0.05$ was set for all analyses.

RESULTS

Phytochemical studies

The phytochemical analysis of the leaf extract by qualitative study showed the presence triterpenoids and flavonoids and the absence of volatile oils, tannins, alkaloids and saponins.

Wound contraction

On day 19, the extract treated animals showed a 98% reduction (Fig. 1) in the wound area as compared with the controls (88%) (Fig. 2) and the difference was statistically significant ($p < 0.05$) (Table 1).

Mean clinical epithelization time

L. camara treated animals were found to epithelize faster (17.20 ± 0.12) when compared with the controls (19.00 ± 0.14) ($p < 0.05$). The complete closure of wounds was observed on day 19 in the control group, while the rats treated with extract took 17 days to heal (Fig. 3).



Figure 1. Experimental wound area before and after treatment on day 19.



Figure 2. Control wound area before and after day 19.

Collagen synthesis

Hydroxyproline content is an index of collagen and measures the synthesis of neo-collagen. Estimation of hydroxyproline content revealed the highest synthesis of collagen in extract treated groups compared with the controls (Fig. 4).

Table 1. Effect of *L. camara* in on wound contraction

Wound area (mm ²)	Placebo control ($n = 6$)	Experimental ($n = 6$)
Day 1	197.00 $\pm$ 0.89	198.60 $\pm$ 2.80
Day 3	189.20 $\pm$ 1.24 (4%)	181.40 $\pm$ 4.90 (8%)
Day 5	163.20 $\pm$ 5.42 (17%)	138.80 $\pm$ 8.39 (30%)
Day 7	145.20 $\pm$ 4.49 (26%)	98.40 $\pm$ 11.02 (50%)
Day 9	118.40 $\pm$ 6.04 (40%)	65.80 $\pm$ 9.02 (66%)
Day 11	86.40 $\pm$ 5.38 (56%)	42.00 $\pm$ 0.00 (78%)
Day 13	53.20 $\pm$ 1.90 (72%)	29.00 $\pm$ 2.90 (85%)
Day 15	36.00 $\pm$ 2.00 (81%)	18.20 $\pm$ 3.80 (90%)
Day 17	28.40 $\pm$ 2.1 (85%)	10.00 $\pm$ 2.50 (94%)
Day 19	22.40 $\pm$ 1.30 (88%)	4.00 $\pm$ 2.09 ^a (98%)

$n = 6$, Values are expressed as mean $\pm$ SEM.

^a $p < 0.002$ versus control.

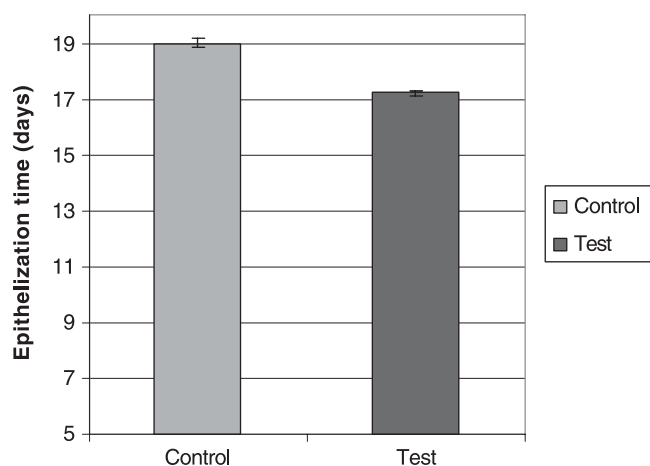


Figure 3. Bar diagram showing the epithelization time of controls and treated group of animals.

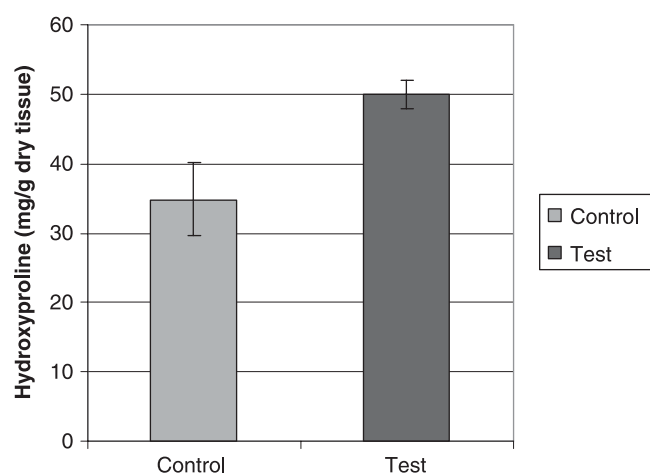


Figure 4. Bar diagram showing hydroxyproline content of the granulation tissue obtained from controls and treated group of animals.

Histological study

The histological study of the granulation tissue obtained on day 10 from the experimental animals showed well organized bands of collagen, fibroblasts and few inflammatory cells (Fig. 5). Whereas the granulation tissue obtained from controls showed disorganized collagen fibres and few fibroblasts (Fig. 6).

Microbiological study

Microbial organisms such as *Staphylococcus aureus*, *Klebsiella pneumoniae* and *Escherichia coli* and were sensitive to the *L. camara* leaf extract. *Salmonella choleraesuis* was resistant to the extract (Table 2).

DISCUSSION

The ethanol extract of *L. camara* increased the rate of wound contraction and the rate of epithelialization in excision wounds. The moderately increased hydroxyproline content of the granulation tissue indicates increased

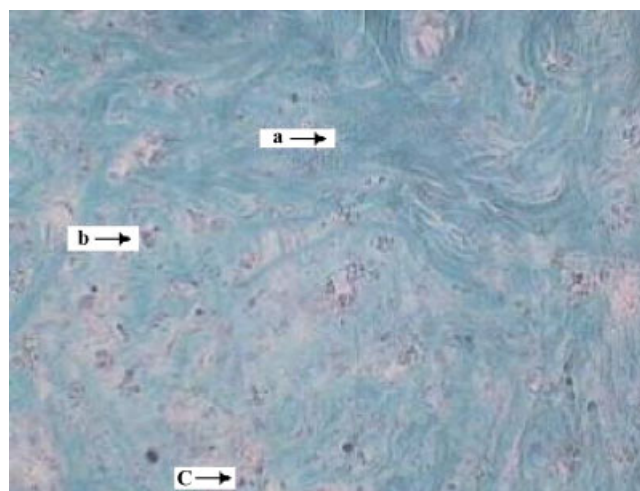


Figure 5. Experimental group granulation tissue from excision wound showing well organized collagen (Masson's trichrome stain) a, collagen fibres; b, fibroblast; c, inflammatory cells.

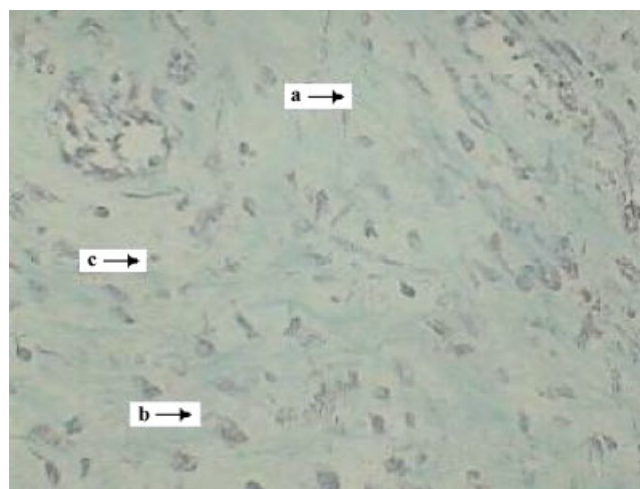


Figure 6. Control group granulation tissue from excision wound model showing disorganized collagen (Masson's trichrome stain) a, collagen fibres; b, fibroblast; c, inflammatory cells.

Table 2. Antimicrobial activity of *L. camara* extract

Bacteria	Turbidity pattern
<i>Escherichia coli</i> (ATCC 5922)	+++
<i>Klebsiella pneumoniae</i> (ATCC 700603)	+++
<i>Staphylococcus aureus</i> (ATC C4827)	+++
<i>Salmonella choleraesuis</i> (ATCC 14028)	+

collagen turnover. Collagen, the major component of granulation tissue, strengthens and supports extracellular tissue which is composed of the amino acid, hydroxyproline, and it has been used as a biochemical marker for tissue collagen (Kumar *et al.*, 2006). The reduced wound area of the excision was due maybe to the antimicrobial effect of the leaf extract which was reported elsewhere (Barre *et al.*, 1997). The extract might have inhibited the growth of microorganisms that infect and delay the healing process. In many rural parts of India

people still apply the *L. camara* leaf extract whenever they get cut wounds.

The phytochemical analysis of the leaf extract by qualitative method showed the presence of triterpenoids and flavanoids and the absence of other constituents. This observation was similar to that reported by Seawright and Hrdlicka (1977) for the presence of triterpene acids such as lantadene A, reduced lantadene A and lantadene B. Flavonoids are excellent antioxidants and free radical scavengers. There is a controversy over the role of free radicals on wound healing (Juranek and Bezek, 2005). However, the wound healing potential of honey was well correlated with its antioxidant properties (Henriques *et al.*, 2006).

Our antimicrobial study with the crude extract evidenced the presence of antimicrobial activity against *Staphylococcus aureus*, *Klebsiella pneumoniae* and *Escherichia coli*. The triterpene content of *L. camara* was reported to have antimicrobial activity against *Staphylococcus aureus* and *Salmonella typhi* (Barre *et al.*, 1997). The antimicrobial activity may explain the wound healing activity of the extract in terms of reduced wound area in the excision wound model.

As *L. camara* was reported to have oral toxicity, liver toxicity and embryotoxicity effects (Sharma *et al.*, 1999),

its effect was studied after topical administration. Moreover, in indigenous practice, people use aqueous extracts of *L. camara* externally to cure skin diseases in the Republic of Benin, without any report of toxicity (de Souza, 1988). The antimicrobial effect of triterpenoids could promote healing in the excision wound model by controlling infection and the antioxidant effect of its flavonoids could promote wound healing by minimizing oxidative damage to the excision wound tissue.

The data demonstrate that the topical administration of *L. camara* may facilitate wound healing. Its wound healing promoting activity in excision wounds could be due to its antioxidant, antimicrobial or some unknown activities. However, it needs to be studied further to isolate the active ingredients that promote wound healing, before considering it for clinical use.

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